📖 1. Room Overview

The **Defensive Security Info** room provides a high-level introduction to the "Blue Team" side of cybersecurity. While offensive security focuses on breaking in, **Defensive Security** focuses on two main goals: **Preventing** intrusions and **Detecting** them when they happen.

🕵️ 2. Task 1: What is Defensive Security?

Defensive security is a multi-layered approach to protecting an organization's digital assets. It involves several specialized roles:

🏢 SOC (Security Operations Center)

The SOC is the "command center" where security analysts monitor the network 24/7.

- **Goal:** To identify suspicious activity before it becomes a breach.
- **Tools:** They use **SIEM** (Security Information and Event Management) tools to aggregate logs from every computer on the network.

🔍 Digital Forensics

When a hack happens, Forensic Analysts investigate the "digital crime scene."

- **Activity:** Analyzing memory (RAM), hard drives, and network logs to see exactly what the hacker did.
- **Evidence:** Maintaining a **Chain of Custody** to ensure evidence is valid in court.

🧪 3. Task 2: Threat Intelligence & Hunting

Defensive security isn't just waiting for an alarm to go off; it's about being proactive.

- **Threat Intel:** Researching known hackers and their "recipes" (TTPs - Tactics, Techniques, and Procedures).
- **Malware Analysis:** Taking a virus apart in a safe "sandbox" to see how it works without letting it infect the real network.
- **Risk Management:** Identifying which assets (like customer data) are most valuable and ensuring they have the strongest locks.

🚨 4. Task 3: Incident Response (IR)

The **Incident Response** process is a structured plan for when things go wrong. Most organizations follow the **NIST** or **SANS** framework:

1. **Preparation:** Training and tools.
2. **Identification:** Detecting the attack.
3. **Containment:** "Unplugging" the infected computer to stop the spread.
4. **Eradication:** Removing the malware or the hacker's access.
5. **Recovery:** Getting systems back online.
6. **Lessons Learned:** Analyzing the attack to prevent it from happening again.

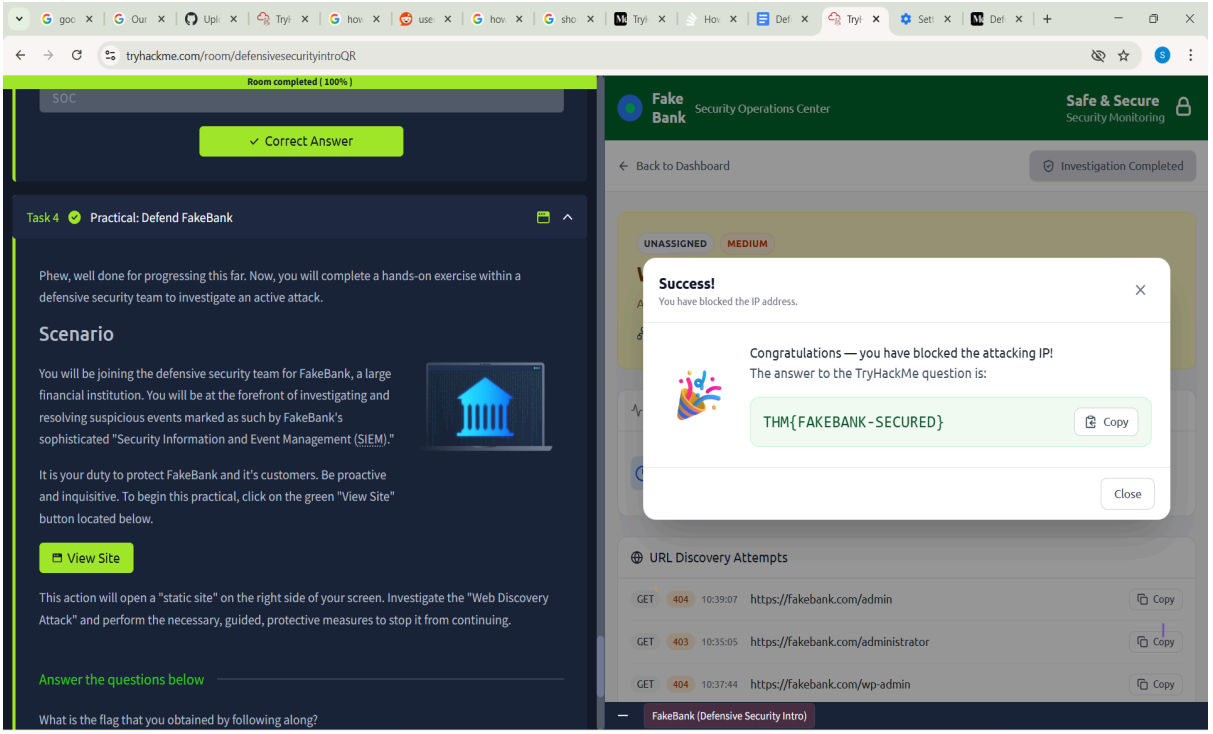
🚩 5. Lab Activity: Security Operations

In this lab, I acted as a SOC Analyst to investigate an alert and mitigate a simulated threat.

ANSWER:

What is the flag that you obtained by following along?

THM[redacted]



🎓 6. Conclusion & Skills Gained

Completing this module has provided a solid foundation in:

- Understanding the **SOC Analyst** workflow.

- Identifying the phases of **Incident Response**.
- Differentiating between **SIEM, EDR, and IDS/IPS** technologies.